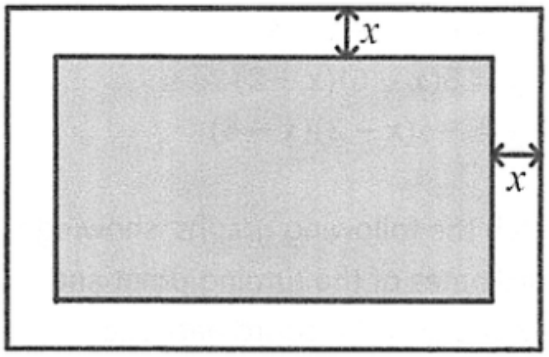
Quadratics: Problem Solving

Questions 1–13

Question 1. A rectangular garden is 30m by 40m. The path surrounding a rectangular pond has a consistent width of x and total area of 864m^2. Find the width and length of the pond.



Question 2. A room is comprised of two areas; one square with length x m, and one rectangular with width $(x+2)$ m and length 10 m. It has a total area of 44 m^2. Find the value of x.

Question 3. For the equations below, without manipulating the equation, state the number of real roots:

a) $y = (x+1)^2 + 2$
b) $y = -(x+1)^2 + 2$
c) $y = (x-4)^2 - 3$
d) $y = -(x-4)^2 - 3$
e) $y = (x+2)^2$

Question 4.

a) Sketch the graph of $y = \frac{3}{4}x(x-1)$ for $0 \leq x \leq 4$.
b) Find the minimum of the curve.

Question 5.

a) Sketch the graph of $y = \frac{22}{25}x(200 - x)$ for $0 \leq x \leq 200$.

b) Find the maximum of the curve.

Question 6. Sketch the graph of $y = a(x - b)^2 + c$ for constants a, b and c in each of the cases below. Label where the curve intersects with the axes.

a) $a > 0, b > 0, c > 0$

b) $a > 0, b < 0, c > 0$

c) $a < 0, b > 0, c < 0$

d) $a < 0, b < 0, c < 0$

Question 7. For the graph of $y = a(x - b)^2 + c$ where a, b and c are constants, state the conditions for the graph to have a minimum in the top left quadrant and the y-intercept to be positive.

Question 8. For the graph of $y = a(x - b)^2 + c$ where a, b and c are constants, state the conditions for the graph to have a maximum in the top right quadrant and the y-intercept to be negative.

Question 9. The graph of $y = x^2 + 3x + k$ where k is a constant has a minimum with coordinates $(-\frac{3}{2}, -\frac{1}{2})$. Find the coordinates of where the curve intersects

a) the y-axis.

b) the x-axis.

Question 10. The graph of $y = -x^2 + 4kx + k$ where k is a constant has a maximum with coordinates $(2, 5)$. Find the coordinates of the points where the curve intersects

c) the y-axis.

d) the x-axis.

Question 11. By completing the square, show that the solutions to $ax^2 + bx + c = 0$ are given by $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Question 12. A curve has equation $y = 2x^2 + x + k$ where $k \neq 0$. Given that the roots of the equation have a difference of 2.5, find the value of k.

Question 13. A curve has equation $y = -5x^2 - kx + 3$ where $k \neq 0$. Given that the roots of the equation have a difference of 1.6, find the possible values of k.